



PATHGENIE AI

Navigate Your Future with AI-Powered Career Insights
and Skill Intelligence.

INTERNSHIP PROJECT REPORT

SUBMITTED FOR THE:

DATA SCIENCE INTERNSHIP

Submitted By:

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CN/INT/2026/409

Department:

Information Technology (IT)

College:

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Internship Organization:

CODE-A-NOVA

Internship Domain:

Data Science

Internship Duration:

2 Months

Submission Year:

May-2026

Abstract:

PathGenie AI is an intelligent career recommendation system developed to help students and beginners identify suitable technology domains based on their skills, interests, and technical knowledge. The application uses Machine Learning techniques to analyse user responses and provide real-time career recommendations in domains such as Web Development, Machine Learning, Python Development, and Cyber Security.

The project is developed using Python and Streamlit with a clean and beginner-friendly interface. It demonstrates the practical implementation of Data Science concepts including data preprocessing, Exploratory Data Analysis (EDA), feature engineering, Machine Learning model training, and intelligent prediction systems while helping users make better career decisions.

Introduction:

In today's rapidly evolving technology industry, students and beginners are exposed to a wide range of career opportunities such as Artificial Intelligence, Data Science, Web Development, Cyber Security, Machine Learning, and Cloud Computing. Due to the increasing number of available domains, many learners often face confusion while selecting the most suitable career path based on their skills, interests, and technical knowledge. To address this problem, **PathGenie AI** was developed as an intelligent career recommendation system that uses Artificial Intelligence and Machine Learning to analyse user responses and predict the most suitable technology domain. The project is developed using Python, Streamlit, Pandas, and Scikit-learn to create a simple, interactive, and user-friendly platform. It also demonstrates practical implementation of important Data Science concepts such as data preprocessing, feature engineering, Machine Learning model training, prediction systems, and interactive web application development. The main objective of this project is to help users make informed career decisions through accurate, data-driven, and personalised career recommendations.

Objectives of the Project:

- ✓ To help students and beginners identify suitable career paths
- ✓ To reduce confusion during career selection
- ✓ To provide AI-powered career recommendations
- ✓ To analyse user skills, interests, and technical knowledge
- ✓ To demonstrate practical implementation of Machine Learning concepts
- ✓ To create a beginner-friendly and interactive platform
- ✓ To develop a real-time prediction and recommendation system

Tech Stack:

Technology	Purpose in Project
Python	Used as the core programming language for application development and Machine Learning implementation.
Streamlit	Used to build the interactive web application interface.
Pandas	Used for data preprocessing, cleaning, and handling datasets.
NumPy	Used for numerical operations and data manipulation.
Scikit-learn	Used to train and evaluate the Machine Learning model.
Joblib	Used to save and load trained Machine Learning models.
Pickle / PKL Models	Used to store trained models and encoders for prediction.

Dataset Description:

The project uses the **Career Recommendation Dataset (career_recommender.csv)**, which contains information related to students' educational background, interests, skills, certifications, and career preferences. The dataset is used to train the Machine Learning model for predicting suitable technology domains based on user responses.

The dataset contains both categorical and numerical data, including features such as gender, undergraduate course, interests, technical skills, CGPA, certification courses, and work experience. These features help the system analyse user profiles and generate intelligent career recommendations.

Before model training, the dataset was preprocessed by handling missing values, removing duplicate records, and encoding categorical variables to improve prediction accuracy and model performance.

Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) was performed to understand the dataset structure, analyse important patterns, and prepare the data for Machine Learning model training. Different visualisations and statistical methods were used to identify relationships between user skills, interests, educational background, and career domains.

Dataset Loading

```
df = pd.read_csv("../dataset/career_recommender.csv")
df.head()
```

	What is your name?	What is your gender?	What was your course in UG?	What is your UG specialization? Major Subject (Eg; Mathematics)	What are your interests?	What are your skills ? (Select multiple if necessary)	What was the average CGPA or Percentage obtained in under graduation?	Did you do any certification courses additionally?	If yes, please specify your certificate course title.	Are you working?	If yes, then what is/was your first Job title in your current field of work? If not applicable, write NA.	Have you done masters after under graduation? If yes, mention your field of masters.(Eg; Masters in Mathematics)
0	A.Uha Priya	Female	B.Sc	Computer Applications	Cloud computing	Python;SQL;Java	85	Yes	Linux,Git	Yes	Software	MCA
1	Aadil	Male	B.E	Computer Science Engineering	Technology	Critical Thinking, Analytic Thinking, SQL, Pro...	66.5	Yes	Microsoft certification	Yes	Computer Software Engineer	NaN
2	Aakriti	Female	BA	Psychology	Understand human behaviour	management;Communication skills	64.6	Yes	Resilience psychology	No	NaN	NaN
3	Aanchal sharma	Female	MBA	Commerce	Sales/Marketing;Trading;Understand human behav...	Accounting Skills;Critical Thinking	75.525	No	No	Yes	Relationships manager	NaN
4	Aangkeeta Sarkar	Female	B.Tech	Instrumentation Engineering	Technology	\nPLC Allen Bradley;PLC Ladder Logic;LabVIEW...	70.68	Yes	Extreme Productivity (Blinkist Summary)	Yes	Plant Instrumentation Engineer	NaN

Dataset Information

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1195 entries, 0 to 1194
Data columns (total 12 columns):
#   Column                                                                                               Non-Null Count  Dtype
---  -
0   What is your name?                                           1195 non-null   object
1   What is your gender?                                         1195 non-null   object
2   What was your course in UG?                                  1195 non-null   object
3   What is your UG specialization? Major Subject (Eg; Mathematics)  1195 non-null   object
4   What are your interests?                                     1195 non-null   object
5   What are your skills ? (Select multiple if necessary)        1194 non-null   object
6   What was the average CGPA or Percentage obtained in under graduation?  1195 non-null   object
7   Did you do any certification courses additionally?          1195 non-null   object
8   If yes, please specify your certificate course title.        1194 non-null   object
9   Are you working?                                             1195 non-null   object
10  If yes, then what is/was your first Job title in your current field of work? If not applicable, write NA.  899 non-null    object
11  Have you done masters after under graduation? If yes, mention your field of masters.(Eg; Masters in Mathematics)  469 non-null    object
dtypes: object(12)
memory usage: 112.2+ KB
```

	What is your name?	What is your gender?	What was your course in UG?	What is your UG specialization? Major Subject (Eg: Mathematics)	What are your interests?	What are your skills ? (Select multiple if necessary)	What was the average CGPA or Percentage obtained in under graduation?	Did you do any certification courses additionally?	If yes, please specify your certificate course title.	Are you working?	If yes, then what is/was your first Job title in your current field of work? If not applicable, write NA.	Have you done masters after undergraduation? If yes, mention your field of masters. (Eg: Masters in Mathematics)
count	1195	1195	1195	1195	1195	1194	1195	1195	1194	1195	899	469
unique	1161	3	59	295	615	1007	301	3	523	2	483	236
top	JITEN KUMAR BARIK	Male	B.Tech	Computer Science Engineering	Technology	NO	76	Yes	No	Yes	Student (Unemployed)	Masters in Business Administration
freq	3	806	488	174	132	44	56	611	580	713	116	42

Data Cleaning

```
df.dropna(inplace=True)
df.drop_duplicates(inplace=True)
```

```
df.isnull().sum().sum()

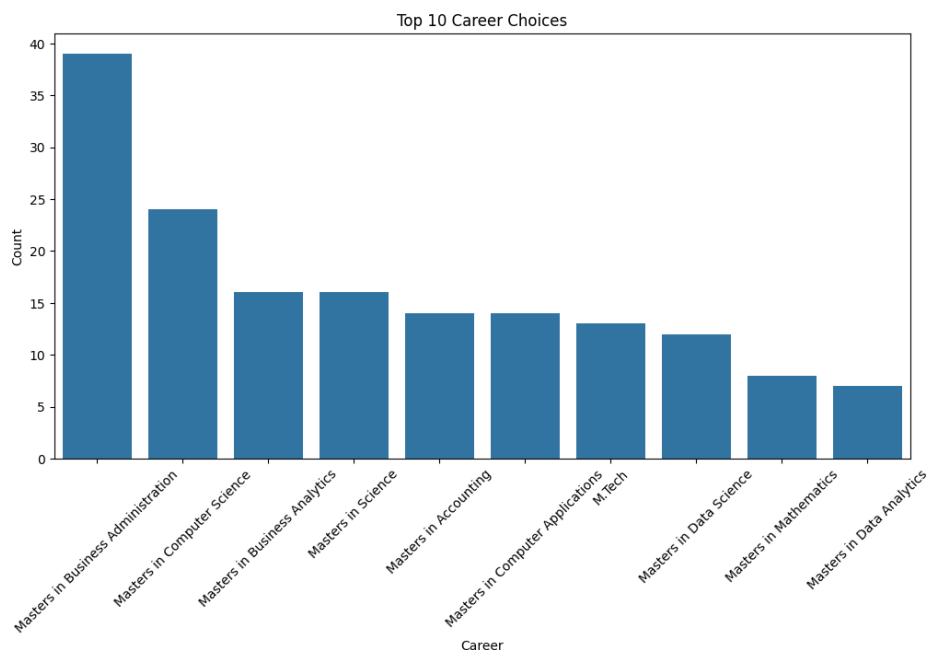
np.int64(0)

df.duplicated().sum()

np.int64(0)
```

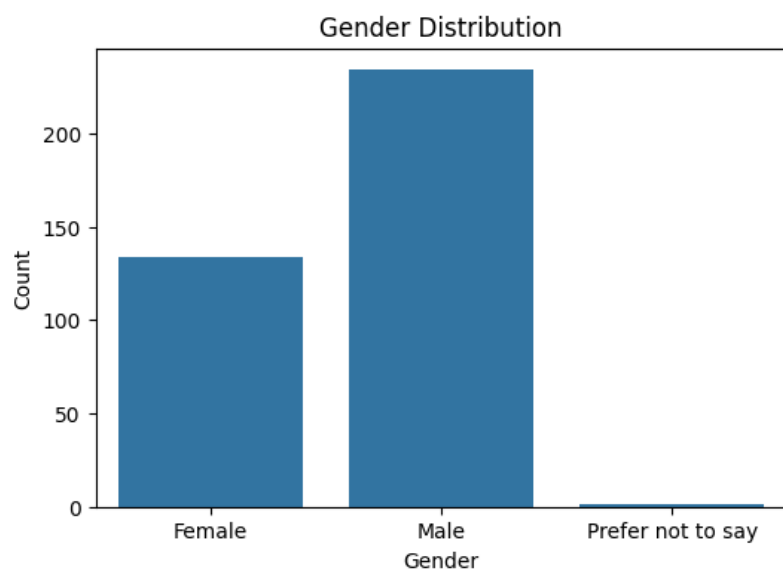
Top Career Choices Visualisation

```
plt.figure(figsize=(12,6))
top_careers = df['career'].value_counts().head(10)
sns.barplot(
    x=top_careers.index,
    y=top_careers.values
)
plt.xticks(rotation=45)
plt.title("Top 10 Career Choices")
plt.xlabel("Career")
plt.ylabel("Count")
plt.show()
```



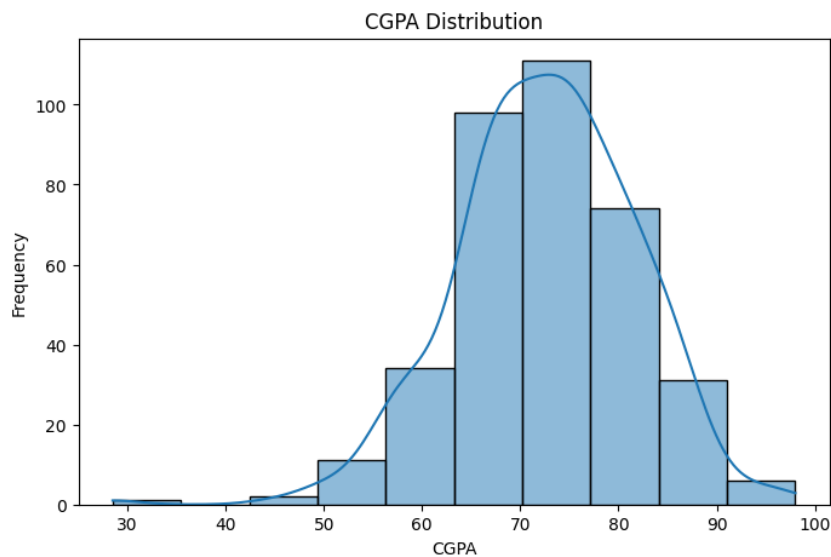
Gender Distribution

```
plt.figure(figsize=(6,4))
sns.countplot(x='gender', data=df)
plt.title("Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.show()
```



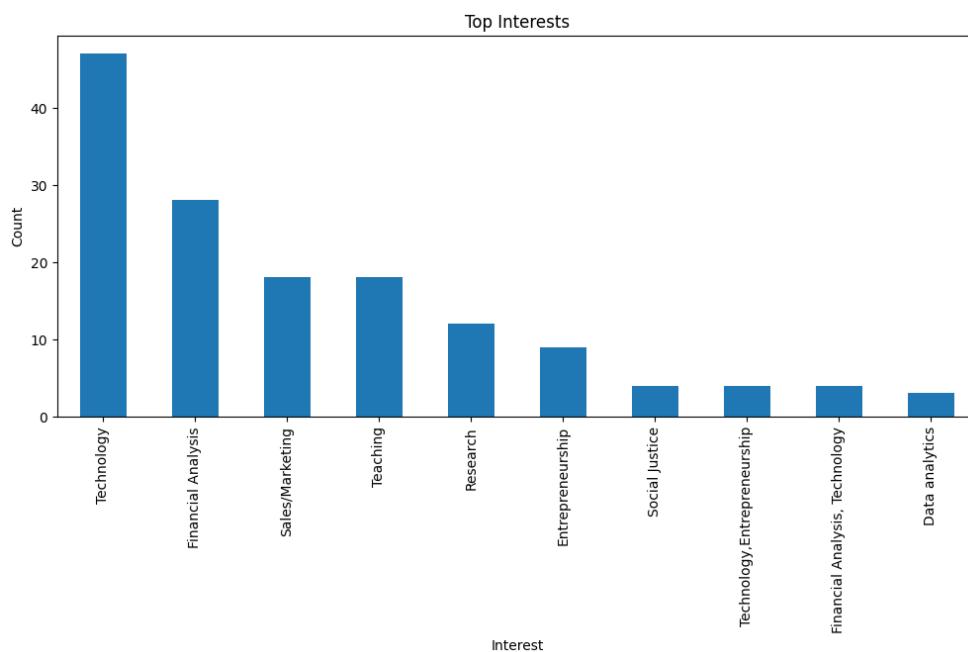
CGPA Distribution

```
plt.figure(figsize=(8,5))
sns.histplot(df['cgpa'], bins=10, kde=True)
plt.title("CGPA Distribution")
plt.xlabel("CGPA")
plt.ylabel("Frequency")
plt.show()
```



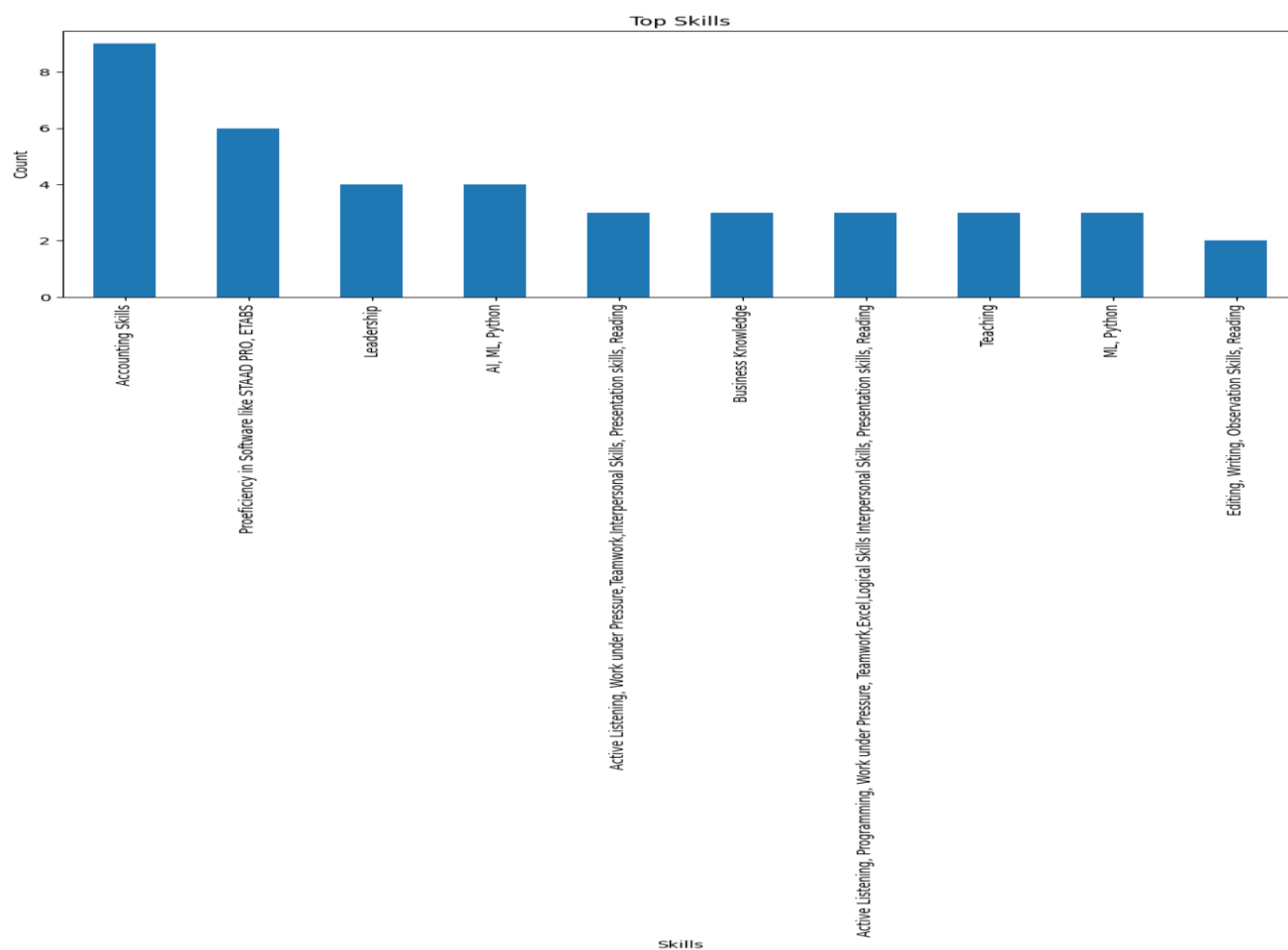
Top Interests Analysis

```
plt.figure(figsize=(12,5))
df['interests'].value_counts().head(10).plot(kind='bar')
plt.title("Top Interests")
plt.xlabel("Interest")
plt.ylabel("Count")
plt.show()
```



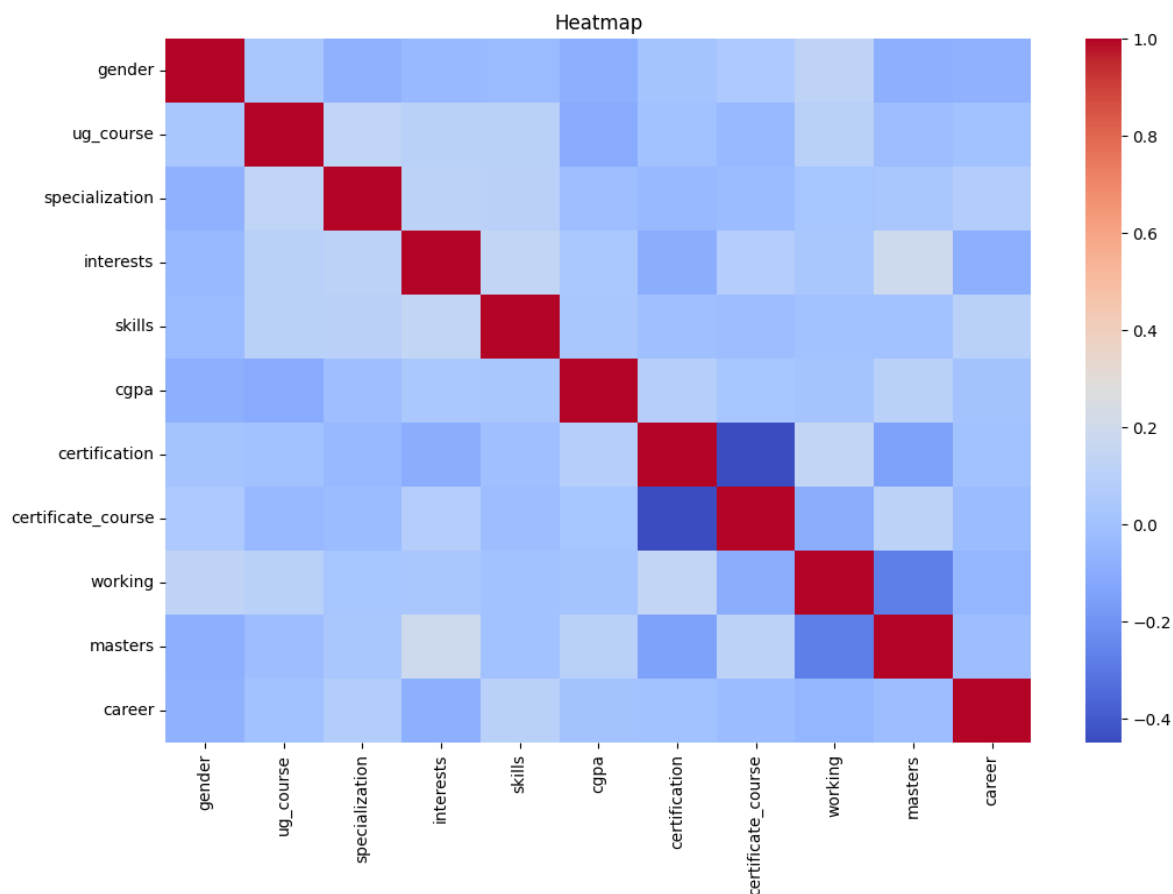
Top Skills Analysis

```
plt.figure(figsize=(12,5))
df['skills'].value_counts().head(10).plot(kind='bar')
plt.title("Top Skills")
plt.xlabel("Skills")
plt.ylabel("Count")
plt.show()
```



Heatmap Generation

```
plt.figure(figsize=(12,8))
sns.heatmap(
    heatmap_df.corr(),
    cmap='coolwarm',
    annot=False
)
plt.title("Heatmap")
plt.show()
```



Machine Learning Workflow:

The Machine Learning workflow implemented in the project includes multiple stages such as preprocessing, feature encoding, model training, evaluation, prediction generation, and model saving.

Feature Encoding

```
from sklearn.preprocessing import LabelEncoder
encoders = {}
for col in df.columns:
    if df[col].dtype == 'object':
        le = LabelEncoder()
        df[col] = le.fit_transform(
            df[col].astype(str)
        )
        encoders[col] = le
```

Saving Encoders

```
import joblib
joblib.dump(
    encoders,
    "../model/label_encoder.pkl"
)
print("All encoders saved successfully!")
```

All encoders saved successfully!

Feature and Train-Test Split

```
X = df.drop('career', axis=1)
y = df['career']
from sklearn.preprocessing import LabelEncoder
career_encoder = LabelEncoder()
y = career_encoder.fit_transform(y)

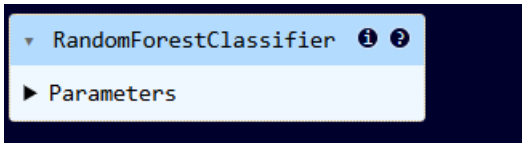
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

Model Used:

The project uses the **Random Forest Classifier** algorithm for career prediction. Random Forest is an ensemble Machine Learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Random Forest Model Training

```
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier(
    n_estimators=200,
    random_state=42
)
model.fit(X_train, y_train)
```



Prediction Generation

```
y_pred = model.predict(X_test)
```

Accuracy Evaluation

```
from sklearn.metrics import accuracy_score
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

Model Evaluation

The trained Random Forest Classifier model was evaluated using classification metrics and prediction analysis techniques. The model produced reliable and consistent prediction results for career recommendation tasks.

The evaluation process included:

- Prediction generation on test data
- Classification report analysis
- Confusion matrix visualisation
- Feature importance analysis

The model showed good performance in analysing user skills and generating suitable career recommendations.

Classification Report

```
from sklearn.metrics import classification_report
print(
    classification_report(
        y_test,
        y_pred
    )
)
```

Classification Report:

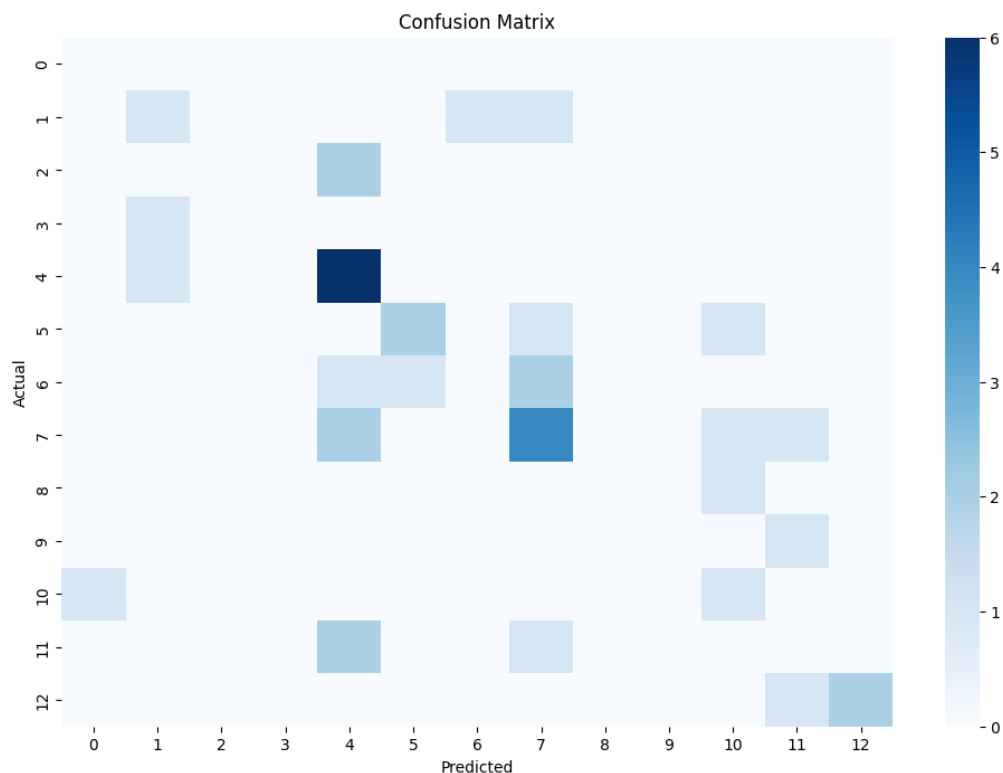
	precision	recall	f1-score	support
2	0.00	0.00	0.00	0
3	0.33	0.33	0.33	3
4	0.00	0.00	0.00	2
5	0.00	0.00	0.00	1
6	0.46	0.86	0.60	7
7	0.67	0.50	0.57	4
8	0.00	0.00	0.00	4
9	0.44	0.50	0.47	8
10	0.00	0.00	0.00	1
11	0.00	0.00	0.00	1
12	0.25	0.50	0.33	2
13	0.00	0.00	0.00	3
14	1.00	0.67	0.80	3
accuracy			0.41	39
macro avg	0.24	0.26	0.24	39
weighted avg	0.36	0.41	0.37	39

Confusion Matrix

```

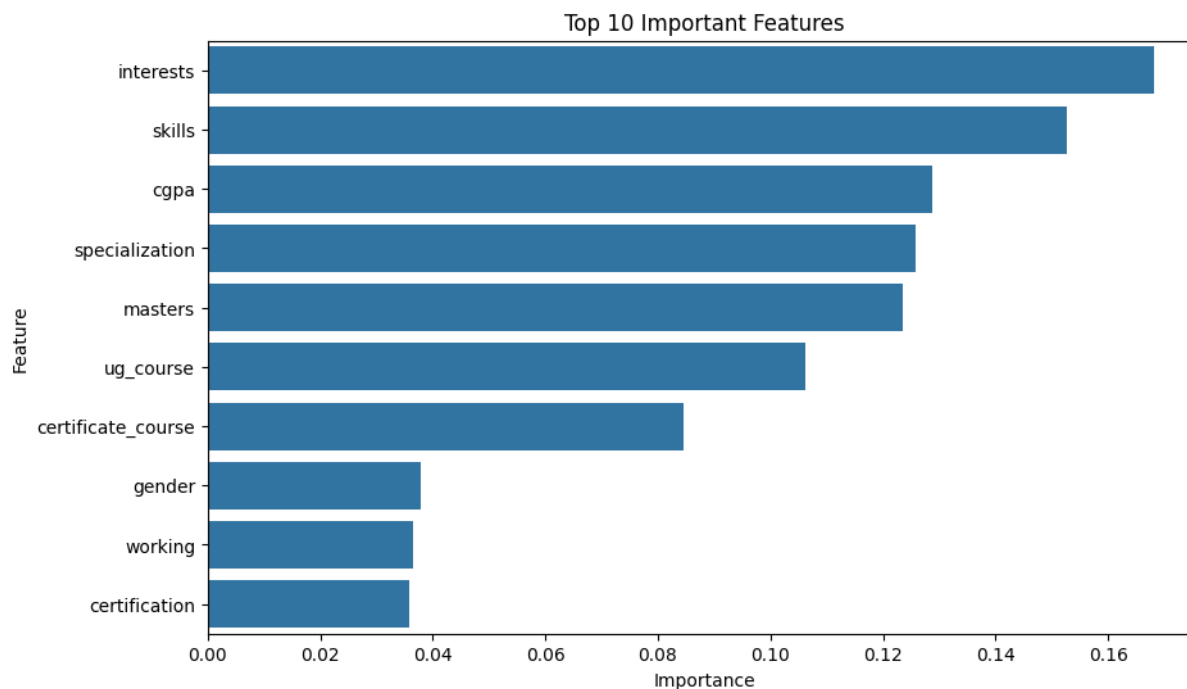
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(
    y_test,
    y_pred
)
plt.figure(figsize=(12,8))
sns.heatmap(
    cm,
    annot=False,
    cmap="Blues"
)
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

```



Feature Importance Visualisation

```
plt.figure(figsize=(10,6))
sns.barplot(
    x='Importance',
    y='Feature',
    data=importance.head(10)
)
plt.title("Top 10 Important Features")
plt.show()
```



Model Saving

```
import joblib
joblib.dump(model, '../model/career_model.pkl')
print("Original model saved successfully!")
```

```
Original model saved successfully!
```

```
feature_columns = X.columns.tolist()
joblib.dump(
feature_columns,
'../model/feature_columns.pkl'
)
```

```
['../model/feature_columns.pkl']
```

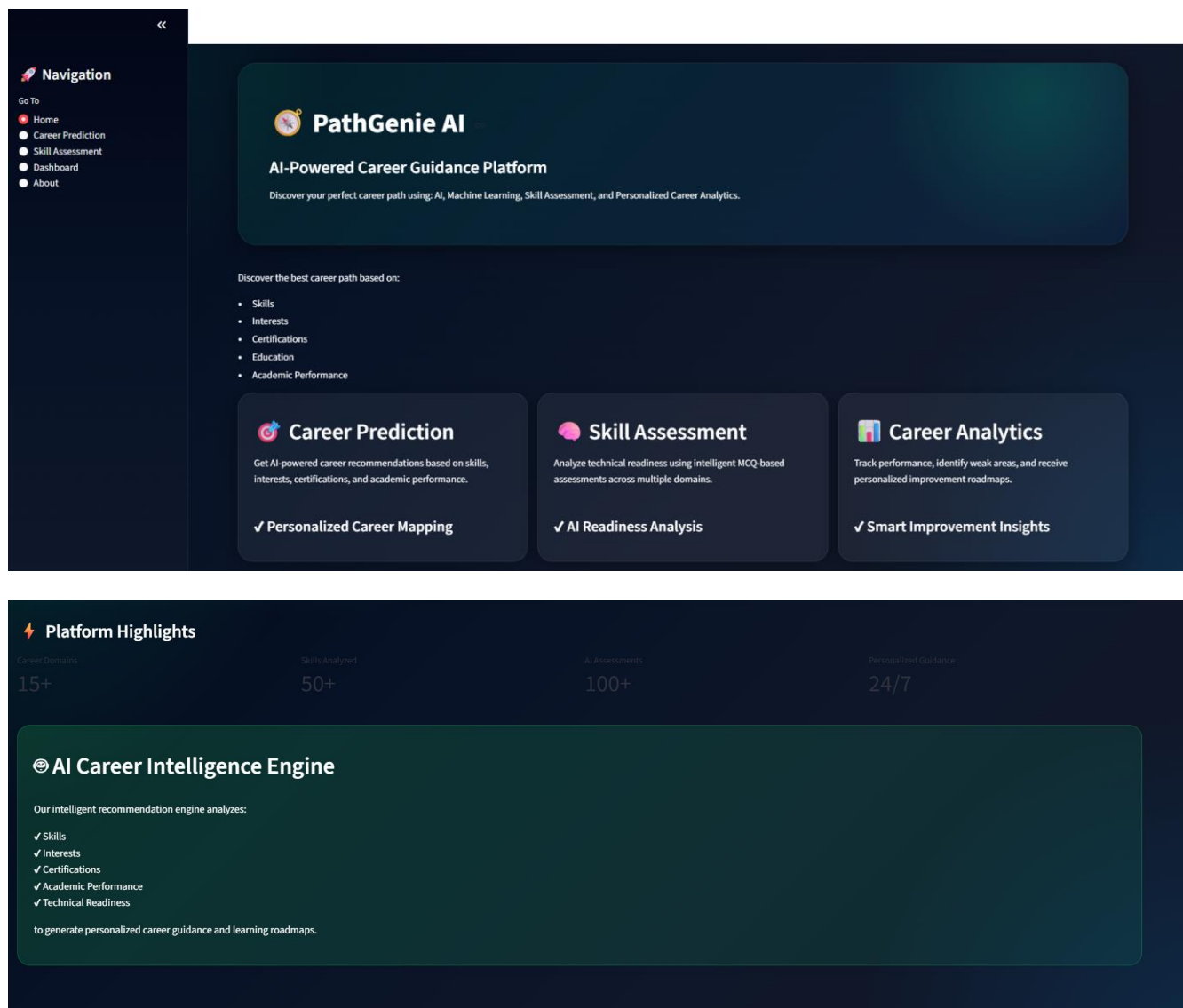
Results and Output:

The developed system successfully predicts suitable career domains based on user skills, interests, and educational background. Users answer interactive questionnaires through the Streamlit web application, and the trained Random Forest model generates intelligent career recommendations in real time.

The application provides a clean and interactive interface for displaying prediction results and guiding users toward suitable technology domains.

Application Screenshots:

Home Page:



Career Prediction



Career Prediction

Fill Your Details

Enter your Name

Select Gender

UG Course

Specialization

CGPA

Enter interests

Select Skills

Any Certification?

Masters Completed?

Certified Course

Currently Working?



Recommended Career: 3



Career Readiness Score: 70%

Skill Assessment:



Skill Assessment

Analyze your technical readiness using intelligent assessments.

Choose Skill

Q1. What is Python?

Choose Answer

- ☒ Programming Language
- ☐ Database
- ☐ Operating System
- ☐ Browser

Q2. Which keyword is used to define a function in Python?

Choose Answer

- ☒ def
- ☐ define
- ☐ add
- ☐ function

Submit Assessment

✓ Your Score: 5/100

✓ Correct Answers: 1/20

✗ Needs Improvement
Focus more on fundamentals, coding practice, and real-world problem solving.

Weak Areas Analysis

Improve: mcq

Improve: code

Improve: scenario

Improve: fill_blank

Weak Areas Analysis

Improve: mcq

Improve: code

Improve: scenario

Improve: fill_blank

Recommended Study Plan

Recommended Daily Practice: 3 Hours/Day

Focus on:

- Fundamentals
- Beginner Projects
- Hands-on Practice

Dashboard:

Personalized Career Dashboard

User Profile

Name

Shiny

Interests

AI, Web Development, Cyber Security

Skills

None selected

Recommended Career

3

Readiness Score

70%

Assessment Analytics

Start Score

Python

Readiness Score

5/100

Current Score

1/20

Career Readiness Meter

Overall Career Readiness: 70%

Weak Areas Analysis

⚠ Improve: mcq

⚠ Improve: code

⚠ Improve: scenario

⚠ Improve: fill_blank

🔍 AI Improvement Suggestions

Career Match

3

Based on your assessment and profile, this career best matches your skills and interests.

Recommended Plan

Recommended Practice Time:

3 Hours/Day

Focus Areas:

- Fundamentals
- Beginner Projects
- Problem Solving

Final AI Insight

You currently need stronger fundamentals for becoming a successful 3.

Focus on skill-building and consistent daily learning.

🔍 AI Career Insights

Best Career Match

3

Your profile strongly aligns with AI and Data Science career paths.

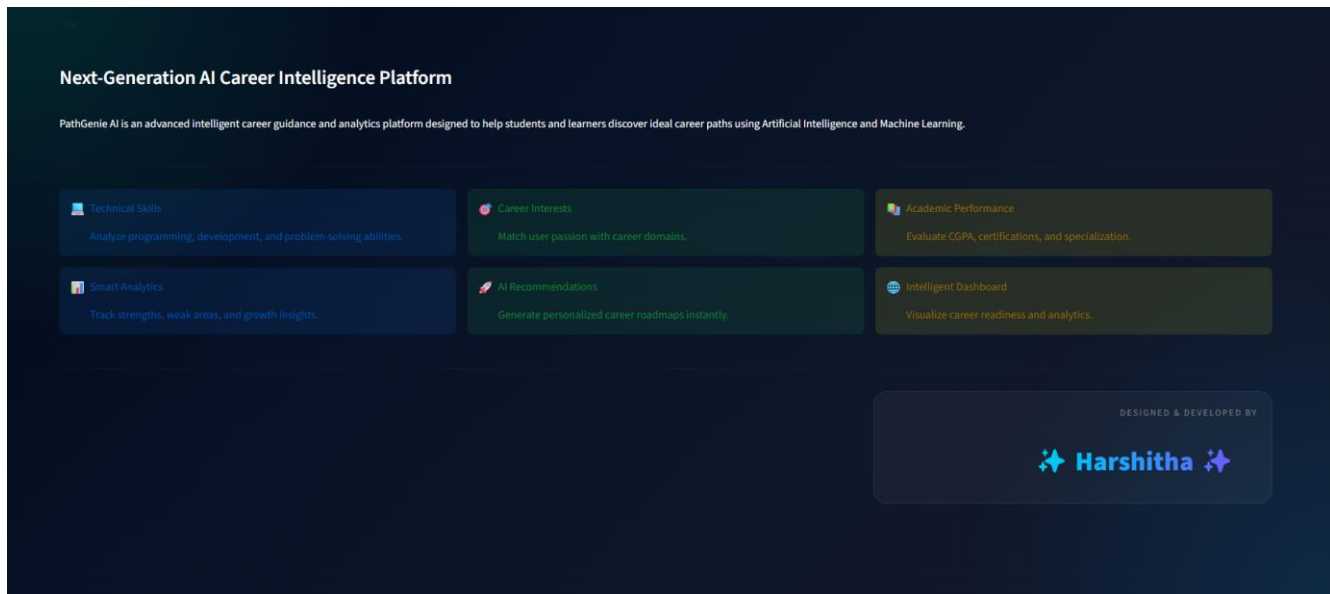
Improvement Suggestion

Focus more on:

- SQL Optimization
- Deep Learning
- DSA Practice

Recommended Practice: 2 Hours/Day

About



Live link:

<https://pathgenie-ai.onrender.com/>

Features of the Application:

- ✓ AI-based career recommendation system
- ✓ Interactive Streamlit web application
- ✓ Real-time career prediction
- ✓ Beginner-friendly interface
- ✓ Skill and interest-based analysis
- ✓ Machine Learning-powered recommendations
- ✓ Fast prediction system using saved models

Future Enhancements:

- ✓ Resume analysis integration
- ✓ Career roadmap generation
- ✓ More technology domain recommendations
- ✓ User authentication system

- ✓ Database integration
- ✓ Cloud deployment

Conclusion:

PathGenie AI successfully demonstrates the practical implementation of Machine Learning and Data Science concepts in a real-world application by combining Exploratory Data Analysis (EDA), data preprocessing, model training, prediction systems, and interactive web application development to create an intelligent career recommendation system. The application helps students and beginners identify suitable technology domains based on their skills and interests while providing a simple, interactive, and beginner-friendly user experience. The project also highlights how Artificial Intelligence can be used to solve real-world career guidance problems through data-driven recommendations.

